Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 27-03–2026

SUBSCRIPTION BASED MEMBERSHIP ENGINE WITH TIERED LOGIC SUBSCRIBTION

SDG: SDG No : 9 – Industry, Innovation, Infrastructure

Presented By:

V NANDA SRINIVAS – VTU25911– Reg.No.23UECS0612

Courses Handling Faculty:

Dr. Hafizur Rahman- Assistant Professor

INTRODUCTION

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- Many organizations use subscription models to provide services such as online learning, streaming, software access, and membership-based platforms.
- In traditional systems, membership details, subscription plans, and payment records are often managed manually or with basic tools.
- There is a need for an automated system that can manage memberships accurately and reduce manual effort.
- This project is important because it helps organizations manage different subscription levels using tiered logic such as Basic, Standard, and Premium plans.
- The system has real-world relevance as many businesses rely on subscription-based services to improve customer engagement and revenue.
- The proposed system allows users to register, log in, select subscription plans, and access features based on their chosen membership level.

PROBLEM STATEMENT

- Many organizations manage membership and subscription details manually or using basic systems.
- Manual handling of user data, plan selection, and payment tracking can lead to errors and inefficiency.
- Existing systems often lack proper automation for managing different subscription tiers and user access levels.
- Difficulty in tracking subscription validity and renewal dates may result in poor user experience.
- Managing multiple membership plans without a structured system can cause data inconsistency and confusion.
- There is a need for a centralized platform that simplifies membership management using tiered logic.
- Therefore, an automated Subscription-Based Membership Engine is required to improve efficiency, accuracy, and service quality.

OBJECTIVE(S)

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- **To design and develop :** A full stack web-based Subscription-Based Membership Engine that allows users to register, log in, select tiered plans (Basic, Standard, Premium), and manage their membership details efficiently.

- **To implement and improve :** Automated tiered logic for controlling user access, managing subscription validity, and securely storing membership data to enhance accuracy, performance, and user experience.

PROPOSED SYSTEM

- The proposed system is a full stack web application that automates the management of subscription-based memberships.
- It allows users to register, log in, and select subscription plans such as Basic, Standard, and Premium based on their requirements.
- The tiered logic controls user access to features depending on the selected membership level. The system securely stores user information, subscription details, and plan validity.
- It reduces manual errors and improves efficiency by automating plan selection, renewal tracking, and data management.
- The application provides a user-friendly interface for both users and administrators to manage memberships easily.
- Compared to traditional systems, it offers better accuracy, faster processing, and improved data organization.
- The system is scalable, reliable and easily used by the user.

PROPOSED SYSTEM ARCHITECTURE

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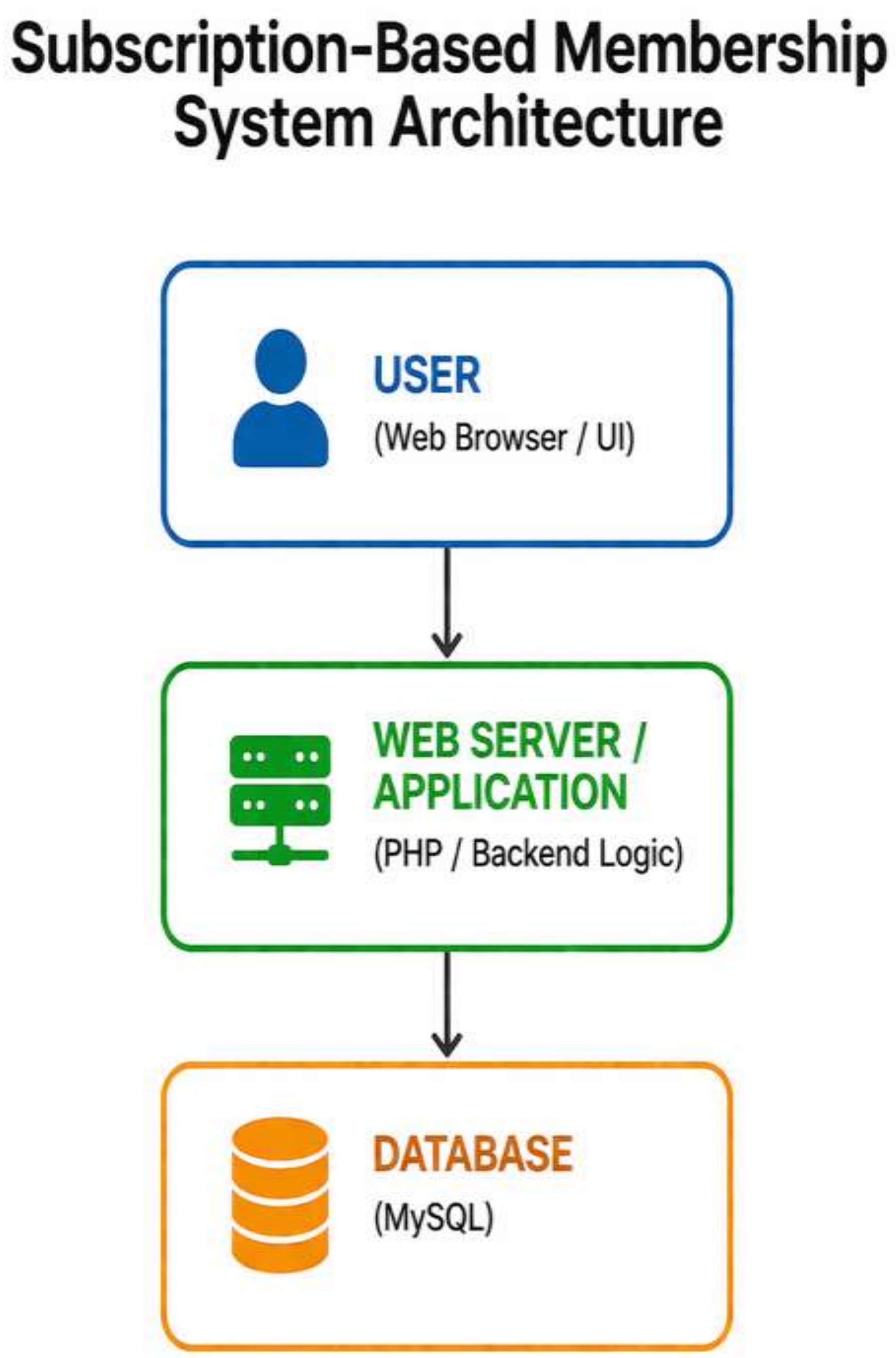
- **Input :**
- Login credentials.
- Selection of subscription plan (Basic, Standard, Premium).
- Payment details.

- **Processing :**
- Validating user details during registration and login.
- Applying tiered logic to determine access level.
- Ensuring secure communication between frontend and data

Output :
- Successful registration or login confirmation.
- Display of available subscription plans.
- Reports or dashboard for admin users.



WORKFLOW

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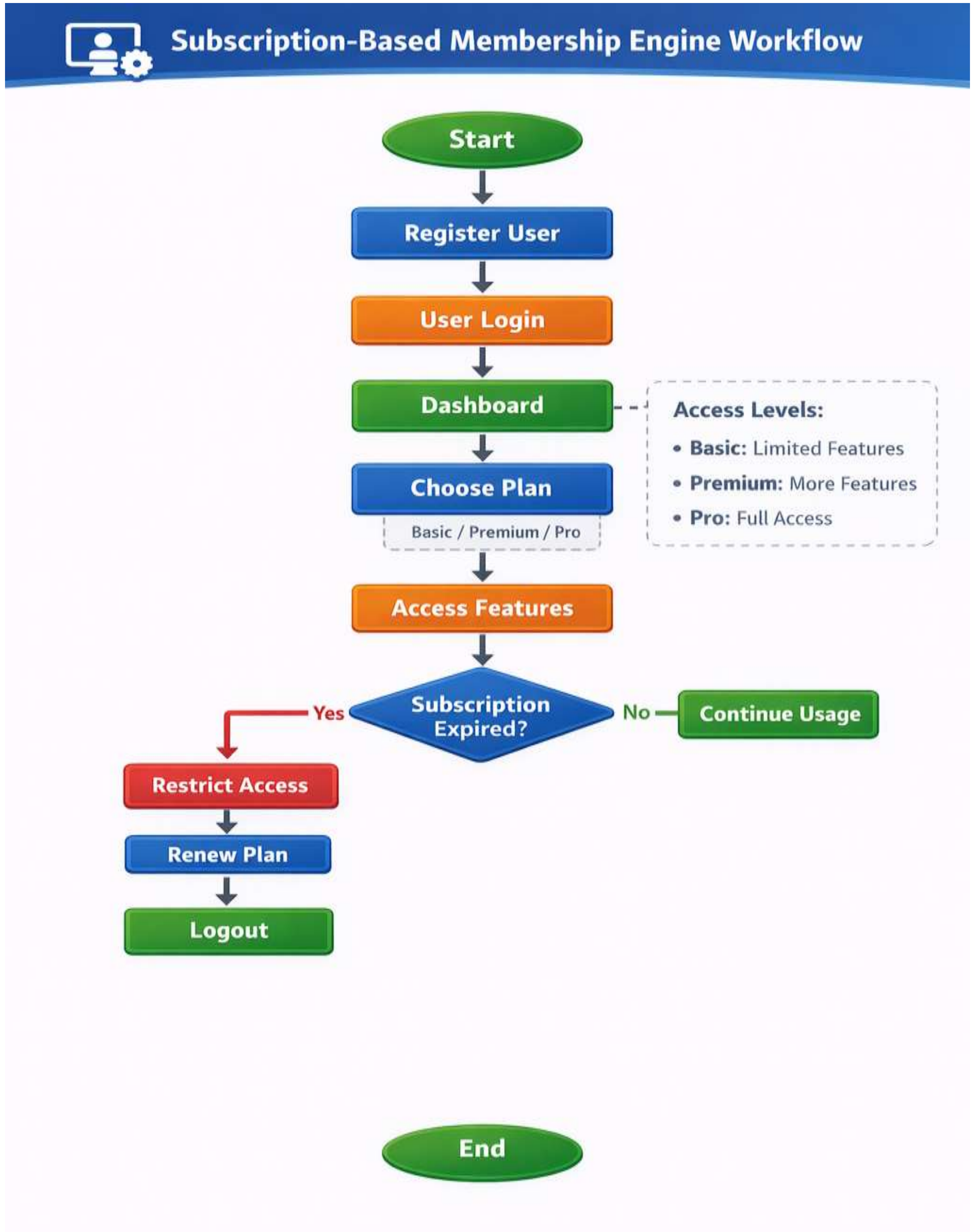

TECHNOLOGIES USED

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Frontend:
- HTML5.
- CSS3.
- JavaScript (ES6).

Backend:
- Java (Spring Boot).

Tools:
- VS Code / Eclipse.
- Postman (API testing).
- Git (version control).
- XAMPP / Tomcat Server.

Hardware Requirements:
- Laptop/Desktop.
- Minimum 4GB RAM & Internet connection.

Software Requirements:
- JDK 8 or MySQL Server.
- Web Browser.
- Operating System (Windows/Linux).

MODULE DESCRIPTION

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Module 1: User Management
This module allows users to create an account, log in, and manage their profile details. It ensures secure authentication so that only registered users can access the system.

Module 2: Core Processing
This module shows different membership plans such as Basic, Standard, and Premium. Users can select or change plans based on their needs, and the system controls access using tiered logic.

Module 3: Database Management
This module stores and manages all data such as user details, subscription plans, and membership validity. It ensures data is stored securely and can be retrieved easily when needed.

Module 4: Reporting & Analytics
This module helps the admin monitor user subscriptions and view important information. It can display simple reports such as number of users and selected membership plans for better management.

IMPLEMENTATION SCREENSHOTS

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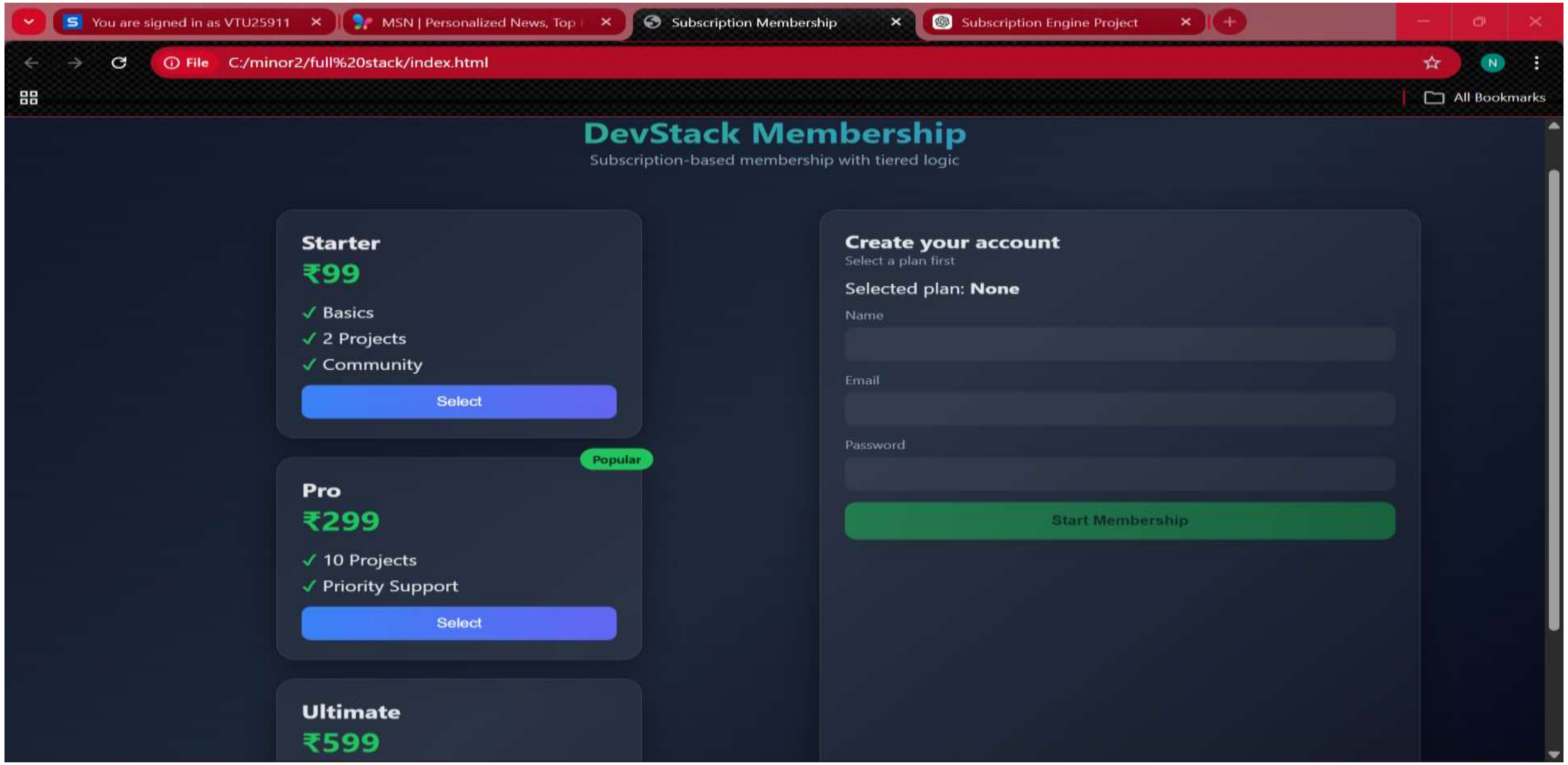
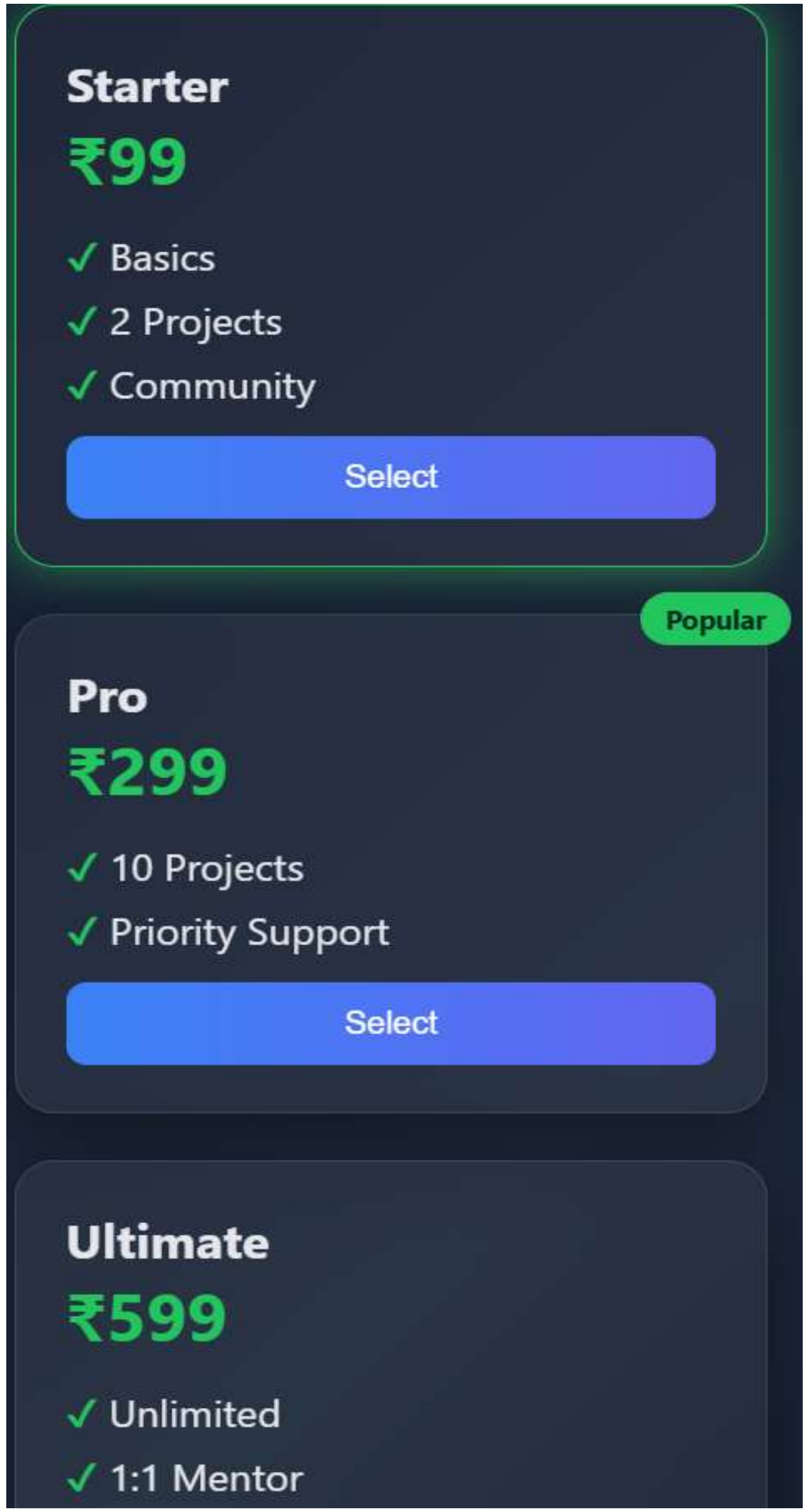


- This is a clean and modern **membership subscription page** where users select a plan and enter their details .
- It provides an interactive UI with tier-based pricing and a subscribe option for dynamic user registration.

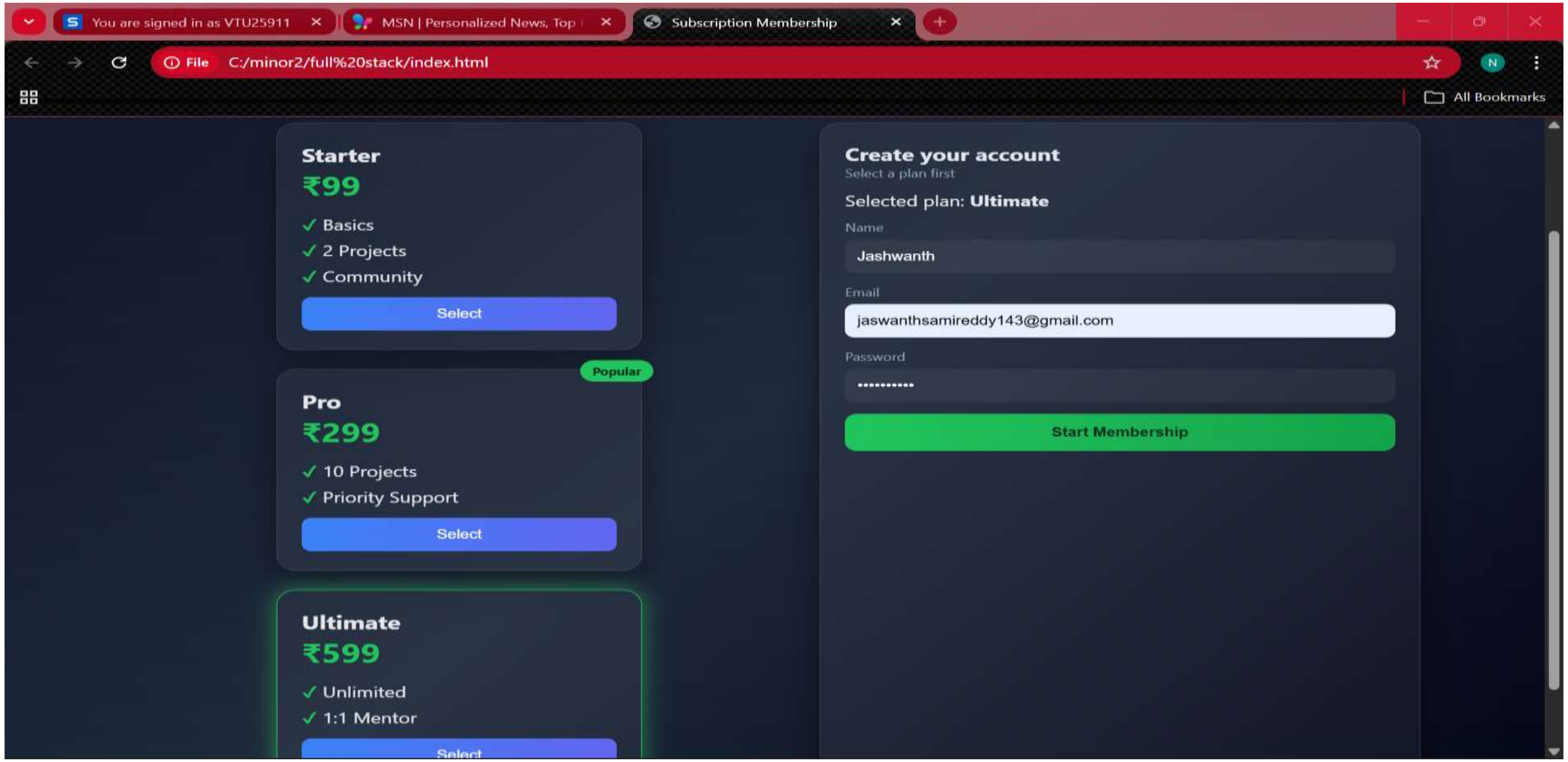
IMPLEMENTATION SCREENSHOTS

- This interface shows a user filling in details and selecting the **Starter, Pro and Ultimate.** And **Pro** is the Popular among those three plans.
- It demonstrates an interactive subscription system where users choose a tier and proceed with registration.
- User should choose any plan among these plans and can go through the User registration.
- Each Plan has different prices and different tiers.



IMPLEMENTATION SCREENSHOTS

- Each Plan has different tiers, prices and reliabilities. So, user can select among them which is perfect for them.
- For User Registration, User needs to enter Name, E-mail and password.
- And can click on the **Start Membership** to continue start their plan.



ADVANTAGES & LIMITATIONS

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ADVANTAGES :

- Reduces manual work by automating membership and subscription management.
- Provides different membership levels (Basic, Standard, Premium) using tiered logic.
- Improves accuracy in storing and managing user data.
- Easy to use interface for both users and admin.
- Saves time by automatically handling registration, login, and plan selection.
- Helps businesses efficiently track subscription validity and renewals.

Limitations:

- Requires internet connection to access the system.
- Initial setup and configuration may take time.
- Security depends on proper implementation of authentication methods.
- Limited features in the basic version of the system.
- Payment integration may need additional configuration.
- System performance depends on server capacity.
- Regular updates may be required to improve functionality.

Thank you